

2018 H2 MATH (9758/01) JC 2 PRELIM
SUGGESTED SOLUTIONS

Qn	Solution
1	Equations and Inequalities
	$\frac{-x^2 + 11x - 11}{x^2 - 4x + 4} + 3$ $= \frac{-x^2 + 11x - 11 + 3(x^2 - 4x + 4)}{x^2 - 4x + 4}$ $= \frac{-x^2 + 11x - 11 + 3x^2 - 12x + 12}{x^2 - 4x + 4}$ $= \frac{2x^2 - x + 1}{x^2 - 4x + 4}$
	$\frac{x^2 - 11x + 11}{x^2 - 4x + 4} < 3$ $\frac{-x^2 + 11x - 11}{x^2 - 4x + 4} + 3 > 0$ $\frac{2x^2 - x + 1}{(x - 2)^2} > 0$ Method 1 Since the discriminant of $2x^2 - x + 1$ is $(-1)^2 - 4(2)(1) = -7 < 0$, and the coefficient of $x^2 = 2$ is positive, $2x^2 - x + 1 > 0$ for all $x \in \mathbb{R}$. Method 2 $2x^2 - x + 1 = 2 \left[\left(x^2 - \frac{1}{2}x + \frac{1}{2} \right) \right] = 2 \left(x - \frac{1}{4} \right)^2 + \frac{7}{8} > 0 \text{ for all } x \in \mathbb{R}.$ Therefore, solving $\frac{1}{(x - 2)^2} > 0$, we have $x \in \mathbb{R}$, $x \neq 2$. $\begin{array}{c} + \qquad \qquad \qquad + \\ \hline \qquad \qquad \oplus \qquad \qquad \qquad \\ \qquad \qquad 2 \end{array}$

Qn	Solution
2	Vectors
(a)	$\mathbf{a} \times \mathbf{b}$ is a vector that is perpendicular to <u>both</u> $\mathbf{a}$ and $\mathbf{b}$. Other possible answers: Normal vector of a plane that is <u>parallel</u> to both $\mathbf{a}$ and $\mathbf{b}$ (no marks for normal vector of a plane containing vectors $\mathbf{a}$ and $\mathbf{b}$)
(b)	Area of triangle OAB $= 0.5 \left(\left \overrightarrow{OA} \right \right) \left(\left \overrightarrow{OB} \right \right) \sin \angle AOB$ $= 0.5 \left \overrightarrow{OA} \times \overrightarrow{OB} \right$
	$\left \overrightarrow{OA} \times \overrightarrow{OB} \right$ is the area of a parallelogram with <u>$\overrightarrow{OA}$ and $\overrightarrow{OB}$ as its adjacent sides</u>. Alternative: $\left \overrightarrow{OA} \times \overrightarrow{OB} \right$ is area of parallelogram $OACB$ (and include a diagram with O, A, a fourth point C, and B, with the 2 parallel sides also indicated clearly in diagram). Cannot accept $\left \overrightarrow{OA} \times \overrightarrow{OB} \right$ is area of sum of two triangles, because question asked for geometrical meaning in relation to an appropriate quadrilateral.

Qn	Solution
3	Complex Numbers
(i)	$1 - e^{in\theta}$ $= e^{\frac{1}{2}in\theta} \left(e^{-\frac{1}{2}in\theta} - e^{\frac{1}{2}in\theta} \right)$ $= e^{\frac{1}{2}in\theta} \left(\cos \frac{n\theta}{2} - i \sin \frac{n\theta}{2} - \left(\cos \frac{n\theta}{2} + i \sin \frac{n\theta}{2} \right) \right)$ $= e^{\frac{1}{2}in\theta} \left(-2i \sin \frac{n\theta}{2} \right)$ $1 + e^{in\theta}$ $= e^{\frac{1}{2}in\theta} \left(e^{-\frac{1}{2}in\theta} + e^{\frac{1}{2}in\theta} \right)$ $= e^{\frac{1}{2}in\theta} \left(\cos \frac{n\theta}{2} - i \sin \frac{n\theta}{2} + \cos \frac{n\theta}{2} + i \sin \frac{n\theta}{2} \right)$ $= e^{\frac{1}{2}in\theta} \left(2 \cos \frac{n\theta}{2} \right)$
(ii)	Method 1 $ 1 - z + z^2 - z^3 $ $= (1 - z) + (1 - z)z^2 $ $= (1 - z)(1 + z^2) $ $= 1 - z 1 + z^2 $ $= \left e^{\frac{1}{2}i\theta} \left(-2i \sin \frac{\theta}{2} \right) \right \left e^{i\theta} (2 \cos \theta) \right $ $= 2 \sin \frac{\theta}{2} (2 \cos \theta) \quad \left(\left e^{\frac{1}{2}i\theta} \right = 1, \left e^{i\theta} \right = 1, \text{ and } 0 \leq \theta \leq \frac{\pi}{2} \Rightarrow \sin \frac{\theta}{2} > 0 \text{ and } \cos \theta > 0 \right)$ $= 4 \sin \frac{\theta}{2} \cos \theta$

Method 2

$$1 - z + z^2 - z^3 = \frac{1 - (-z)^4}{1 - (-z)} = \frac{1 - z^4}{1 + z} = \frac{1 - e^{i4\theta}}{1 + e^{i\theta}}$$

Using (i),

$$1 - z + z^2 - z^3$$

$$= \frac{1 - e^{i4\theta}}{1 + e^{i\theta}}$$

$$= \frac{e^{i2\theta}(-2i \sin 2\theta)}{e^{i\frac{\theta}{2}} \left(2 \cos \frac{\theta}{2} \right)}$$

$$= \frac{e^{i\frac{3\theta}{2}}(-2i \sin \theta \cos \theta)}{\left(\cos \frac{\theta}{2} \right)}$$

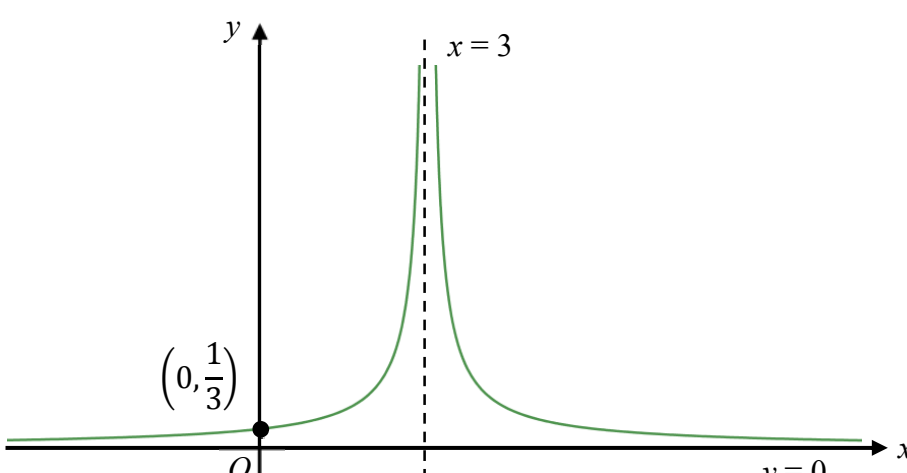
$$= \frac{e^{i\frac{3\theta}{2}} \left(-4i \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \theta \right)}{\left(\cos \frac{\theta}{2} \right)}$$

$$= e^{i\frac{3\theta}{2}} \left(-4i \sin \frac{\theta}{2} \cos \theta \right)$$

Since $\left| e^{i\frac{3\theta}{2}} \right| = 1$, and $0 \leq \theta \leq \frac{\pi}{2} \Rightarrow \sin \frac{\theta}{2} > 0$ and $\cos \theta > 0$

$$\left| 1 - z + z^2 - z^3 \right| = 4 \sin \frac{\theta}{2} \cos \theta \text{ (shown).}$$

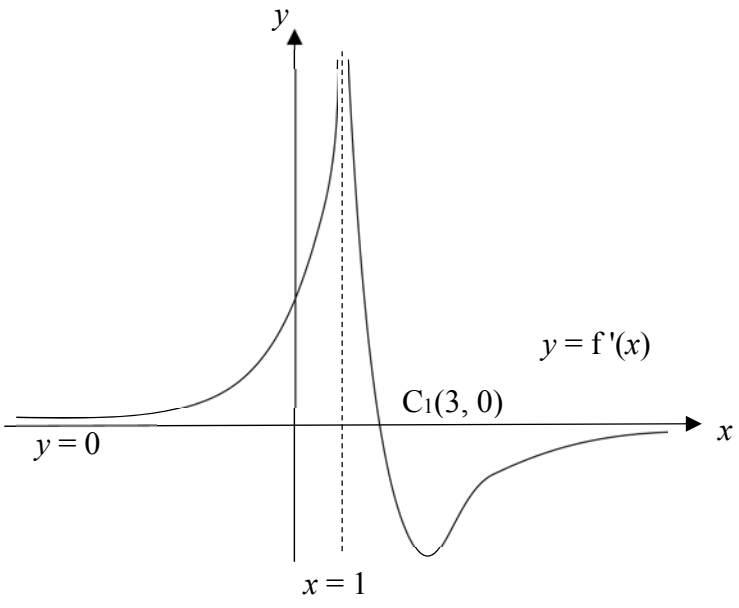
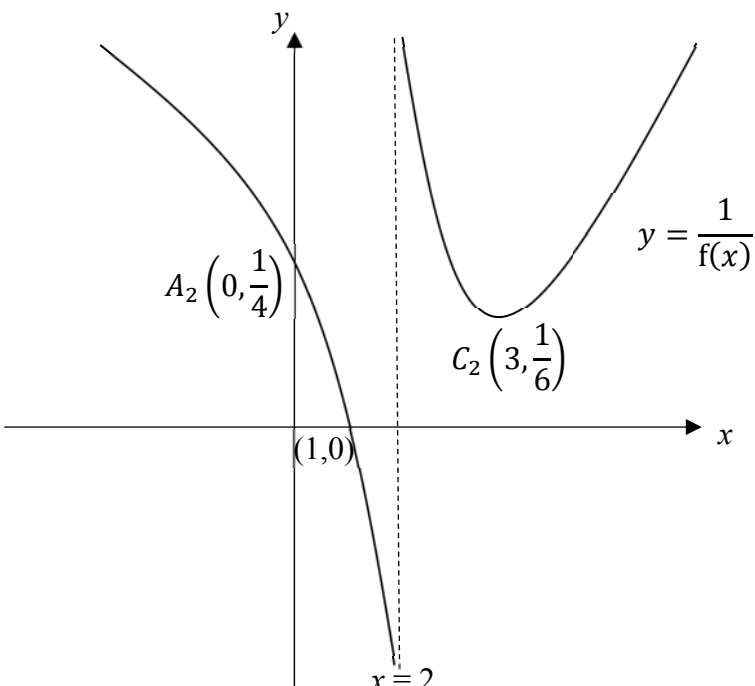
Qn	Solution
4	Techniques of integration
(a)	$\int x\sqrt{5-x^2} \, dx = -\frac{1}{2} \int -2x\sqrt{5-x^2} \, dx$ $= -\frac{1}{2} \frac{(5-x^2)^{\frac{3}{2}}}{\frac{3}{2}} + c$ $= -\frac{1}{3} (5-x^2)^{\frac{3}{2}} + c$
(b)	$\int \sin(\ln x) \, dx$ $= \int (1)(\sin(\ln x)) \, dx$ $= x \sin(\ln x) - \int x \cos(\ln x) \frac{1}{x} \, dx$ $= x \sin(\ln x) - \int \cos(\ln x) \, dx$ $= x \sin(\ln x) - \left[x \cos(\ln x) + \int x \sin(\ln x) \frac{1}{x} \, dx \right]$ $= x \sin(\ln x) - x \cos(\ln x) - \int \sin(\ln x) \, dx$ $\Rightarrow 2 \int \sin(\ln x) \, dx = x \sin(\ln x) - x \cos(\ln x)$ $\Rightarrow \int \sin(\ln x) \, dx = \frac{1}{2} [x \sin(\ln x) - x \cos(\ln x)] + c$
(c)	$\int_{-\frac{\pi}{4}}^{\frac{\pi}{12}} \cos x \sin x \, dx$ $= \int_{-\frac{\pi}{4}}^0 \cos x (-\sin x) \, dx + \int_0^{\frac{\pi}{12}} \cos x (\sin x) \, dx$ $= -\frac{1}{2} \int_{-\frac{\pi}{4}}^0 2 \sin x \cos x \, dx + \frac{1}{2} \int_0^{\frac{\pi}{12}} 2 \sin x \cos x \, dx$ $= -\frac{1}{2} \int_{-\frac{\pi}{4}}^0 \sin 2x \, dx + \frac{1}{2} \int_0^{\frac{\pi}{12}} \sin 2x \, dx$ $= -\frac{1}{2} \left[\frac{-\cos 2x}{2} \right]_{-\frac{\pi}{4}}^0 + \frac{1}{2} \left[\frac{-\cos 2x}{2} \right]_0^{\frac{\pi}{12}}$ $= -\frac{1}{4} (-1 - 0) + \frac{1}{4} \left(-\frac{\sqrt{3}}{2} + 1 \right)$ $= \frac{1}{2} - \frac{\sqrt{3}}{8}$ Recall that $f(x) = \begin{cases} -f(x) & \text{when } f(x) < 0 \\ f(x) & \text{when } f(x) \geq 0 \end{cases}$ Hence $\sin x = \begin{cases} -\sin x & \text{when } -\frac{\pi}{4} \leq x < 0 \\ \sin x & \text{when } 0 \leq x \leq \frac{\pi}{12} \end{cases}$

Qn	Solution
5	Function
(i)	Since $R_f = (0, \infty) \subseteq D_g = (0, \infty)$, gf exists. $gf : x \mapsto \ln \left \frac{1}{x-3} \right$, where $x \in \mathbb{R}, x \neq 3$
(ii)	 Since $y = 3$ cuts the graph of f twice, f is not one-one. Hence f^{-1} does not exist.

(iii)	Maximum $k = 3$ For $\frac{1}{x-3} < 0$, Let $y = -\frac{1}{x-3}$ $x-3 = -\frac{1}{y}$ $x = 3 - \frac{1}{y}$ $\therefore f^{-1}(x) = 3 - \frac{1}{x}$ $D_{f^{-1}} = R_f = (0, \infty)$ $f^{-1} : x \mapsto 3 - \frac{1}{x}, x \in \mathbb{R}, x > 0$
(iv)	The graph of $y = f(x)$ is a reflection of the graph of $y = f^{-1}(x)$ in the line $y = x$.

Qn	Solution
6	Binomial Theorem and Maclaurin's Series

(a) (i)	$y = [1 + \ln(1 + 3x)]^{-\frac{1}{2}}$ $= \left[1 + (3x) - \frac{(3x)^2}{2} + \dots \right]^{-\frac{1}{2}}$ $\approx \left(1 + (3x - \frac{9}{2}x^2) \right)^{-\frac{1}{2}}$ $= 1 + \left(-\frac{1}{2} \right) \left(3x - \frac{9}{2}x^2 \right) + \frac{\left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right)}{2} \left(3x - \frac{9}{2}x^2 \right)^2 + \dots$ $= 1 - \frac{3}{2}x + \frac{9}{4}x^2 + \frac{3}{8}(9x^2) + \dots$ $= 1 - \frac{3}{2}x + \frac{45}{8}x^2 + \dots$
(a) (ii)	$\int_0^1 y \, dx \approx \int_0^1 1 - \frac{3}{2}x + \frac{45}{8}x^2 \, dx$ $= \frac{17}{8}$ Approximation is not good as $x=1$ is not close to 0 OR the two graphs deviate significantly in the interval $[0,1]$.
(a) (iii)	$y = 1 - \frac{3}{2}x$
(b)	$\frac{\cos(2x)}{1 - \sin x}$ $\approx \frac{1 - \frac{1}{2}(2x)^2}{1 - x}$ $= \frac{1 - 2x^2}{1 - x}$ $= (1 - 2x^2)(1 - x)^{-1}$ $= (1 - 2x^2)(1 + x + x^2) + \dots$ $= 1 + x - x^2 + \dots$ $a = 1, b = 1, c = -1$

Qn	Solution
7	Graphing Techniques - Transformations
(a) (i)	 The graph shows a function $y = f'(x)$ on a Cartesian coordinate system. The x-axis is labeled x and the y-axis is labeled y. A horizontal line $y = 0$ is shown. A vertical dashed line is at $x = 1$. The curve $y = f'(x)$ crosses the x-axis at $C_1(3, 0)$. The curve has a vertical asymptote at $x = 1$.
(a) (ii)	 The graph shows a function $y = \frac{1}{f(x)}$ on a Cartesian coordinate system. The x-axis is labeled x and the y-axis is labeled y. A vertical dashed line is at $x = 2$. The curve $y = \frac{1}{f(x)}$ has a vertical asymptote at $x = 2$. The curve passes through points $A_2(0, \frac{1}{4})$ and $C_2(3, \frac{1}{6})$. The curve also passes through $(1, 0)$.

(b)

$$y = \frac{3x^2 - 6x}{x-1} = 3(x-1) - \frac{3}{x-1}$$

$$\therefore A = 3, B = -3$$

A sequence of transformation from the graph of $y = \frac{1}{x} - x$ to the graph of $y = \frac{3x^2 - 6x}{x-1}$ is:

1. a. **Reflect** graph **about** the x -axis (Replace y with $(-y)$, get $y = x - \frac{1}{x}$)

OR

- b. **reflect** graph **about** the y -axis (Replace x with $(-x)$, get $y = x - \frac{1}{x}$)

2. **Translate** graph 1 unit in the positive x -direction (Replace x with $(x-1)$, get

$$y = (x-1) - \frac{1}{x-1})$$

3. **Scaling** parallel to y -axis by **factor** of 3 (Replace y with $\frac{y}{3}$, get $y = 3(x-1) - \frac{3}{x-1}$)

Other possible transformations: 2, 1a, 3 or 3,1,2 or 2, 3, 1a

Qn	Solution
8	Vectors 3
(i)	$\vec{OA} = \begin{pmatrix} -2 \\ -7 \\ 3 \end{pmatrix} \text{ and } \vec{OB} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ $\vec{AB} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} -2 \\ -7 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix}$ $l: \mathbf{r} = \begin{pmatrix} -2 \\ -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix}, \lambda \in \mathbb{R}$
(ii)	Let θ be the acute angle between l and p. $\sin \theta = \frac{\left \begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right }{\sqrt{1^2 + 4^2 + (-1)^2} \sqrt{1^2 + (-1)^2}}$ $= \frac{ 1 - 4 }{\sqrt{18} \sqrt{2}} = \frac{3}{6}$ $\Rightarrow \theta = 30^\circ$
(iii)	$p: \mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 5$ Subst. $\vec{OA}$ into p, $\text{LHS} = \begin{pmatrix} -2 \\ -7 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = -2 + 7 = 5 = \text{RHS}$ Hence A lies on p. Note: $\angle BAC = 2(30^\circ) = 60^\circ$ and $BA = CA$ (since reflection) ABC is an <u>equilateral</u> triangle.
(iv)	Let F be the foot of perpendicular from B to p $\vec{OF} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} \mu \\ 1 - \mu \\ 1 \end{pmatrix}, \text{ for some } \mu \in \mathbb{R}$ $\begin{pmatrix} \mu \\ 1 - \mu \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 5$ $\Rightarrow \mu - 1 + \mu = 5$ $\Rightarrow \mu = 3$ $\therefore \vec{OF} = \begin{pmatrix} 3 \\ 1 - 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$

$$\vec{AF} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} -2 \\ -7 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \\ -2 \end{pmatrix}$$

By ratio theorem,

$$\vec{AC} = 2\vec{AF} - \vec{AB}$$

$$= 2 \begin{pmatrix} 5 \\ 5 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ 8 \\ -2 \end{pmatrix} = \begin{pmatrix} 8 \\ 2 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix}$$

$\therefore$ Equation of reflection of line l in p is

$$\mathbf{r} = \begin{pmatrix} -2 \\ -7 \\ 3 \end{pmatrix} + k \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix}, k \in \mathbb{R}$$

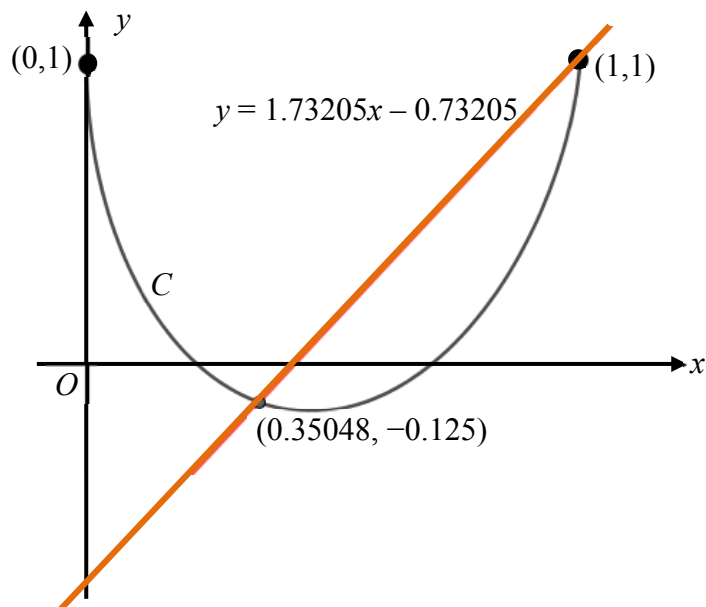
Qn	Solution
9	Differential Equations
(i)	$\frac{dx}{dt} = k(6x - x^2)$ $= k[-(x^2 - 6x)]$ $= k[-(x-3)^2 + 9]$ $= k[9 - (x-3)^2]$ $\int \frac{1}{3^2 - (x-3)^2} dx = \int k dt$ $\frac{1}{2(3)} \ln \left \frac{3+(x-3)}{3-(x-3)} \right = kt + c$ $\ln \left \frac{x}{6-x} \right = 6kt + 6c$ $\frac{x}{6-x} = Ae^{6kt} \quad \text{where } A = \pm e^{6c}$ When $t = 0$, $x = 12$ $\frac{12}{6-12} = Ae^{6(0)}$ $A = -2$ $\frac{x}{6-x} = -2e^{6kt}$ When $t = \frac{1}{12}$, $x = 8.6$ $\frac{8.6}{6-8.6} = -2e^{6k\left(\frac{1}{12}\right)}$ $-\frac{43}{13} = -2e^{0.5k}$ $\ln \frac{43}{26} = 0.5k$ $k = 2 \ln \frac{43}{26}$ $\frac{x}{6-x} = -2e^{12t \ln \frac{43}{26}}$ $x = -2e^{12t \ln \frac{43}{26}} (6-x)$ $x - 2e^{12t \ln \frac{43}{26}} x = -12e^{12t \ln \frac{43}{26}}$ $x = \frac{-12e^{12t \ln \frac{43}{26}}}{1 - 2e^{12t \ln \frac{43}{26}}}$ $= \frac{12e^{12t \ln \frac{43}{26}}}{2e^{12t \ln \frac{43}{26}} - 1}$

(ii)	When $x = 0.75(12) = 9$, $\frac{9}{6-9} = -2e^{12t \ln \frac{43}{26}}$ $\frac{3}{2} = e^{12t \ln \frac{43}{26}}$ $\ln \frac{3}{2} = 12t \ln \frac{43}{26}$ $t = 0.0672 \text{ years (to 3 s.f.)}$
(iii)	$x = \frac{-12e^{6.0372t}}{1 - 2e^{6.0372t}}$ $= \frac{-12}{\frac{1}{e^{6.0372t}} - 2}$ As $t \rightarrow \infty$, $\frac{1}{e^{6.0372t}} \rightarrow 0$, $x \rightarrow 6$ In the long run, the population of the bluegill fish in the lake will <u>decrease and stabilise</u> at 6000. The assumption is that there are no other external factors such as pollution or diseases that may affect the population of bluegill fish in the lake.

Qn	Solution
10	APGP
(a)(i)	$u_n = S_n - S_{n-1}$ $= kn^2 - 3n - [k(n-1)^2 - 3(n-1)]$ $= 2kn - k - 3$ $u_n - u_{n-1} = 2kn - k - 3 - [2k(n-1) - k - 3]$ $= 2k, \text{ a constant}$ $\Rightarrow \{u_n\} \text{ is in AP.}$
(ii)	$\frac{u_3}{u_2} = \frac{u_6}{u_3}$ $(u_3)^2 = (u_2)(u_6)$ $[2k(3) - k - 3]^2 = [2k(2) - k - 3][2k(6) - k - 3]$ $(5k - 3)^2 = (3k - 3)(11k - 3)$ $8k^2 - 12k = 0$ $4k(2k - 3) = 0$ $k = 0 \text{ (rej since } k \neq 0) \text{ or } k = \frac{3}{2}$
(b)(i)	Distance travelled by white-tailed deer, $S_n = \frac{9(1 - 0.95^n)}{1 - 0.95}$ $= 180(1 - 0.95^n)$ As $n \rightarrow \infty$, $S_n \rightarrow 180$. $\therefore$ max distance is 180 m.
(ii)	Distance travelled by jaguar after n leaps, $S_n = \frac{n}{2}[2(6) + (n-1)(-0.1)]$ For jaguar to catch white-tailed deer within n leaps, $\text{Let } D = \frac{n}{2}[12 + (n-1)(-0.1)] - 180(1 - 0.95^n) - 11 \geq 0$ When $n = 49$, $D = -0.21 < 0$ When $n = 50$, $D = 0.3501 > 0$ $\therefore$ min $n = 50$

Qn	Solutions												
11	Differentiation + Integration												
(i)	$x = 1 - \cos^3 \theta, \quad y = 1 - 3 \sin \theta \cos^2 \theta$ $\frac{dx}{d\theta} = -3 \cos^2 \theta (-\sin \theta) = 3 \cos^2 \theta \sin \theta$ $\frac{dy}{d\theta} = -[\sin \theta 6 \cos \theta (-\sin \theta) + 3 \cos^2 \theta (\cos \theta)]$ $= 6 \sin^2 \theta \cos \theta - 3 \cos^3 \theta$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{6 \sin^2 \theta \cos \theta - 3 \cos^3 \theta}{3 \cos^2 \theta \sin \theta}$ $= 2 \tan \theta - \cot \theta \text{ (shown)}$												
(ii)	$2 \tan \theta - \cot \theta = 0$ $2 \tan \theta = \frac{1}{\tan \theta}$ $2 \tan^2 \theta = 1$ $\tan \theta = \pm \frac{1}{\sqrt{2}}$ $\tan \theta = \frac{1}{\sqrt{2}} \left(\text{Q } \tan \theta \text{ is positive as } 0 \leq \theta \leq \frac{\pi}{2} \right)$ $\theta = \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$ Since $\tan \theta = \frac{1}{\sqrt{2}}$ and $0 \leq \theta \leq \frac{\pi}{2}$, $\sqrt{1^2 + (\sqrt{2})^2} = \sqrt{3}$ $\tan \theta = \frac{1}{\sqrt{2}}, \therefore \cos \theta = \frac{\sqrt{2}}{\sqrt{3}} \text{ and } \sin \theta = \frac{1}{\sqrt{3}}$ $\cos \theta = \frac{\sqrt{2}}{\sqrt{3}} \text{ and } \sin \theta = \frac{1}{\sqrt{3}}$ $\therefore x = 1 - \cos^3 \theta = 1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3,$ $y = 1 - 3 \sin \theta \cos^2 \theta = 1 - 3 \left(\frac{1}{\sqrt{3}} \right) \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^2 = 1 - \frac{2}{\sqrt{3}}$ To show minimum point <table><tr><td>x</td><td>$\left(1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3 \right)^-$</td><td>$1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3$</td><td>$\left(1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3 \right)^+$</td></tr><tr><td>$\frac{dy}{dx}$</td><td>-</td><td>0</td><td>+</td></tr><tr><td></td><td></td><td></td><td></td></tr></table> $\therefore \text{At } \left(1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3, 1 - \frac{2}{\sqrt{3}} \right), \text{ it is a minimum point.}$	x	$\left(1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3 \right)^-$	$1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3$	$\left(1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3 \right)^+$	$\frac{dy}{dx}$	-	0	+				
x	$\left(1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3 \right)^-$	$1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3$	$\left(1 - \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^3 \right)^+$										
$\frac{dy}{dx}$	-	0	+										
	Alternative:												

	$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$ $= \left[\frac{d}{d\theta} \left(\frac{dy}{dx} \right) \right] \left(\frac{d\theta}{dx} \right)$ $= \frac{d}{d\theta} (2 \tan \theta - \cot \theta) \frac{1}{3 \cos^2 \theta \sin \theta}$ $= (2 \sec^2 \theta + \operatorname{cosec}^2 \theta) \frac{1}{3 \cos^2 \theta \sin \theta}$ Using GC, $\left[\frac{d}{d\theta} (2 \tan \theta - \cot \theta) \right]_{\tan^{-1} \left(\frac{1}{\sqrt{2}} \right)} \left[\frac{1}{3 \cos^2 \theta \sin \theta} \right]_{\tan^{-1} \left(\frac{1}{\sqrt{2}} \right)}$ $= 5.19616 > 0$ $\therefore$ At $\left(1 - \left(\sqrt{\frac{2}{3}} \right)^3, 1 - \frac{2}{\sqrt{3}} \right)$, it is a minimum point.
(iii)	The gradient of normal is $-\frac{1}{2 \tan \theta - \cot \theta} \bigg _{\frac{\pi}{6}} = -\frac{1}{0.577351} = 1.73205$ Equation of normal is $y - (-0.125) = 1.73205(x - (0.35048))$ $y = 1.73205x - 0.73205$



Coordinates of point on C at $\theta = \frac{\pi}{6}$ is $(0.35048, -0.125)$

Area of area of the region bounded by C , the y -axis and the equation of the normal to the curve at $\theta = \frac{\pi}{6}$

$$\begin{aligned}
 &= \int_{-0.125}^1 x \, dy + \text{area of triangle} \\
 &= \int_{\frac{\pi}{6}}^0 (1 - \cos^3 \theta) (6 \sin^2 \theta \cos \theta - 3 \cos^3 \theta) \, d\theta \\
 &\quad + \frac{1}{2} (0.73205 - 0.125) (0.35048) \\
 &= 0.0997667 + 0.106379 \\
 &= 0.206
 \end{aligned}$$

Alternative:

$$\begin{aligned}
 &= \int_0^{\frac{\pi}{6}} (1 - 3 \sin \theta \cos^2 \theta) (3 \cos^2 \theta \sin \theta) \, d\theta \\
 &\quad - \int_0^{0.35048} (1.73205x - 0.73205) \, dx \\
 &= 0.206
 \end{aligned}$$